A small minimal resolution over the Steenrod algebra at $p = 3$, computed by Gröbner bases

Trace of the `groebner` crate (branch `groebner_lin`)

Abstract

We follow, one reduction at a time, the computation of the minimal free resolution $\cdots \to C_3 \to C_2 \to C_1 \to C_0 \to \mathbb{F}_3$ of the ground field over the mod 3 Steenrod algebra $\mathcal{A}$ by the `groebner` crate, a port of Weinan Lin's Gröbner basis method extended to odd primes. The range is homological degree $s \le 3$ and stem $t - s \le 10$, so internal degrees $t \le 13$. Compared with $p = 2$, there are three new features: the exterior generators Q_k, coefficients in $\mathbb{F}_3$, and polynomial generators of height 3 in the associated graded algebra. Every formula below is taken from the program's own data: the critical pairs it formed, the elements it reduced, and the bases and differentials it produced. Only the bookkeeping of the reduction steps was replayed outside the program, and the replay was checked to reproduce the program's results exactly.

1 Setting

The algebra. We work in the Milnor basis $Q(E)\,\mathcal{P}(R) = Q_0^{\epsilon_0} Q_1^{\epsilon_1} \cdots \mathcal{P}(r_1, r_2, \dots)$ of $\mathcal{A}$, writing $\mathcal{P}^r = \mathcal{P}(r)$. Here $Q_0 = \beta$ is the Bockstein, $|Q_k| = 2 \cdot 3^k - 1$, and $|\mathcal{P}(R)| = \sum_n 2r_n(3^n - 1)$. Thus $|Q_0| = 1$, $|Q_1| = 5$, $|\mathcal{P}^1| = 4$, and $|\mathcal{P}(0,1)| = 16$. Write $r_n = \sum_i d_{i,n} 3^i$ with ternary digits $d_{i,n} \in \{0,1,2\}$. The digit records the power of $\xi_n^{3^i}$ in the dual basis, of degree $2 \cdot 3^i(3^n - 1)$. We record $Q(E)\,\mathcal{P}(R)$ by the monomial

$$Q(E) \prod x_{in}^{d_{i,n}}$$

in the associated graded algebra of the May filtration

$$E^0\mathcal{A} = E[Q_0, Q_1, \dots] \otimes \bigotimes_{i \ge 0,\, n \ge 1} \mathbb{F}_3[x_{in}]/(x_{in}^3).$$

For example $\mathcal{P}^2 \leftrightarrow x_{01}^2$, $\mathcal{P}^3 \leftrightarrow x_{11}$ and $Q_0\mathcal{P}^1 \leftrightarrow Q_0 x_{01}$. A monomial is given by a set of Q_k and a digit $0 \le d \le 2$ in each position. Divisibility is inclusion of the sets together with digitwise $\le$; quotients, greatest common divisors and least common multiples are digitwise too. The product in $E^0\mathcal{A}$ is non-zero exactly when no Q_k repeats and no digit sum reaches 3. Even then it carries a coefficient: a binomial coefficient for each digit, and a sign from reordering the Q_k. The positions in this range, in increasing degree, are

$$x_{01}\ (4), \quad x_{11}\ (12), \quad x_{02}\ (16).$$

The monomial order. A term of a free module is $c\,Q(E)\mathcal{P}(R)\,g$ with $c \in \mathbb{F}_3^\times$ and g a generator. Terms are compared by

1. the generator: the generator added later (of higher degree) comes first;
2. the May weight $\sum_{k \in E}(2k+1) + \sum d_{i,n}(2n-1)$: smaller comes first;

3. the exterior part, read as a binary number with Q_0 most significant: smaller comes first;

4. the digits, read as a number with the position of lowest degree most significant: smaller comes first.

The *leading term* $\mathrm{LT}(x)$ is the first term in this order, and elements are always printed with it first. The leading term of a product $Q(E)\mathcal{P}(R)\cdot x$ is the product of $Q(E)\,x^R$ with $\mathrm{LT}(x)$ in $E^0\mathcal{A}$, *including its coefficient*, whenever that product is non-zero. Elements of a Gröbner basis are normalized to leading coefficient 1. Since $p=3$, coefficients are printed as ± 1.

Names. The generator of C_0 is ι. The generators of C_s, $s\geq 1$, are named after the classes of $\mathrm{Ext}^{s,t}_{\mathcal{A}}(\mathbb{F}_3,\mathbb{F}_3)$ they represent:

- $[a_0]\in C_1$ (dual to Q_0) and $[h_0]\in C_1$ (dual to $\mathcal{P}^1$);
- the tower $[a_0^s]$;
- $[a_1]$ and $[b_0]$ in C_2;
- $[a_0b_0]$ in C_3.

These are only labels. The program numbers the generators of each C_s in the order they are found.

2 The algorithm

The algorithm is the same as at $p=2$. We recall it briefly, with the changes for odd primes.

Levels. Level s is a Gröbner basis $b_0^s, b_1^s, \dots$ of the graph of the differential,

$$\Gamma_s=\{(dy,y) : y\in C_{s+1}\}\subseteq C_s\oplus C_{s+1},$$

for the order on the first component. Each $b_k^s=(x_1 \mid x_2)$ satisfies $dx_2=x_1$, its leading term is that of x_1, and its leading coefficient is 1. The module being resolved enters as "level -1": $\mathbb{F}_3=\mathcal{A}/\mathcal{A}\{Q_0,\mathcal{P}^1,\mathcal{P}^3,\mathcal{P}^9,\dots\}$, so the relations are $Q_0\iota=0$ and $\mathcal{P}^{3^i}\iota=0$.

Critical pairs and singles. When an element with leading term L joins level s, the program forms

- a *critical pair* with each earlier element b whose leading term L' lies on the same generator. Let $M=\mathrm{lcm}(L,L')$ in $E^0\mathcal{A}$, with cofactors $u=M/L'$ and $u'=M/L$. The products $u\cdot b$ and $u'\cdot b_{\mathrm{new}}$ have leading terms λM and μM. The S-element is $\lambda\,u'\cdot b_{\mathrm{new}}-\mu\,u\cdot b$, in which the leading terms cancel. The pairs are pruned by the Gebauer–Möller criteria;
- a *critical single* $m\cdot b_{\mathrm{new}}$ for each generator m of the annihilator of L in $E^0\mathcal{A}$. These are $m=x_{in}^{3-d}$ for each digit d of L in position x_{in}, and $m=Q_k$ for each Q_k in L. Then $m\cdot L=0$ in $E^0\mathcal{A}$, so the product must be reduced.

One degree. In degree t, the program does the following:

1. Every level hands out its pairs and singles of degree t, and the relations of degree t are taken up.
2. Each of them is reduced. The first component is reduced modulo level s: a term $c\,m$ divisible by $\mathrm{LT}(b)$, with quotient u, is cancelled by adding $-c/\lambda$ times $u \cdot b$, where λ is the leading coefficient of $u \cdot b$. The second component is carried along. Then the second component is reduced modulo the first components of level $s+1$, i.e. modulo the boundaries known so far. If the first component has become zero, the second component $z \in C_{s+1}$ is a cycle.
3. Going down from the top level, the results are compared with the elements found earlier in the same degree, normalized, and sorted. A non-zero first component gives a **new basis element**. If both components are zero, the reduction was **useless**. Otherwise z is a cycle that is not a boundary, and it becomes d of a **new generator** of C_{s+2}, or the outcome is **beyond** if $s+2 > 3$.

Truncation. We want stems ≤ 10. A cycle found at level s in degree t is a generator of C_{s+2} in stem $t-s-2$, so level s is only needed for $t \leq s+12$. Relations are only needed for $t \leq 11$, which leaves out $\mathcal{P}^3\iota$ in degree 12 (it would give h_1 in stem 11).

Running it. The trace was produced by a small driver around the public API of the crate. It runs `Resolution::sphere` at $p = 3$ with `SteenrodKind::Classical` and `Bounds { s_max: 3, n_max: 10 }`, calls `compute_through(t)` for $t = 1, \ldots, 13$, and records every reduction through a `PairObserver`.

3 Step by step

Degree $t = 1$

The work of this degree: the relation $Q_0\iota$. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Relation $Q_0\iota = 0$. The module relation enters as the element $(0 \mid Q_0\iota)$ of level -1, so its C_0 component is a candidate for d of a new generator. It is reduced modulo the image $\mathrm{im}(d\colon C_1 \to C_0)$, i.e. the first components of level 0. No term of it is divisible by a leading term of level 0 from earlier degrees. **Outcome: new generator.** The relation $Q_0\iota$ is not a boundary, so C_1 gets a generator $[a_0]$ in degree 1 with $d[a_0] = Q_0\iota$. Level 0 gets the element $b_0^0 = (Q_0\iota \mid [a_0])$, with leading term $Q_0\,\iota$. *End of degree 1.* New basis elements: level 0: b_0^0. New generators: $[a_0] \in C_1$.

Degree $t = 2$

The work of this degree: level 0: single. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Level 0, single $Q_0 \cdot b_0^0$. In $E^0\mathcal{A}$ we have $Q_0^2 = 0$, so the leading term of $Q_0 \cdot b_0^0$ is not the product of the leading terms, and the product has to be reduced:

$$(0 \mid Q_0[a_0]).$$

The first component is zero, so the second component $z = Q_0[a_0]$ is a cycle in C_1. No term of it is divisible by a leading term of level 1 from earlier degrees. The cycle $Q_0[a_0]$ is set aside: cycles are tested once all results of level 0 in this degree are sorted.

The cycles of level 0. There is one. **Outcome: new generator.** The cycle $Q_0[a_0]$ is not a boundary, so C_2 gets a generator $[a_0^2]$ in degree 2 with $d[a_0^2] = Q_0[a_0]$. Level 1 gets the element $b_0^1 = (Q_0[a_0] \mid [a_0^2])$, with leading term $Q_0\,[a_0]$. *End of degree 2.* New basis elements: level 1: b_0^1. New generators: $[a_0^2] \in C_2$.

Degree $t = 3$

The work of this degree: level 1: single. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Level 1, single $Q_0 \cdot b_0^1$. In $E^0\mathcal{A}$ we have $Q_0^2 = 0$, so the leading term of $Q_0 \cdot b_0^1$ is not the product of the leading terms, and the product has to be reduced:

$$(0 \mid Q_0[a_0^2]).$$

The first component is zero, so the second component $z = Q_0[a_0^2]$ is a cycle in C_2. No term of it is divisible by a leading term of level 2 from earlier degrees. The cycle $Q_0[a_0^2]$ is set aside: cycles are tested once all results of level 1 in this degree are sorted.

The cycles of level 1. There is one. **Outcome: new generator.** The cycle $Q_0[a_0^2]$ is not a boundary, so C_3 gets a generator $[a_0^3]$ in degree 3 with $d[a_0^3] = Q_0[a_0^2]$. Level 2 gets the element $b_0^2 = (Q_0[a_0^2] \mid [a_0^3])$, with leading term $Q_0\,[a_0^2]$. *End of degree 3.* New basis elements: level 2: b_0^2. New generators: $[a_0^3] \in C_3$.

Degree $t = 4$

The work of this degree: the relation $\mathcal{P}^1\iota$; level 2: single. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Level 2, single $Q_0 \cdot b_0^2$. In $E^0\mathcal{A}$ we have $Q_0^2 = 0$, so the leading term of $Q_0 \cdot b_0^2$ is not the product of the leading terms, and the product has to be reduced:

$$(0 \mid Q_0[a_0^3]).$$

The first component is zero, so the second component $z = Q_0[a_0^3]$ is a cycle in C_3. There is no level 3 (we stop at $s = 3$). The cycle $Q_0[a_0^3]$ is set aside: cycles are tested once all results of level 2 in this degree are sorted.

The cycles of level 2. There is one. **Outcome: beyond.** The cycle $Q_0[a_0^3]$ would give a generator of C_4, beyond $s = 3$, so it is dropped.

Relation $\mathcal{P}^1\iota = 0$. The module relation enters as the element $(0 \mid \mathcal{P}^1\iota)$ of level -1, so its C_0 component is a candidate for d of a new generator. It is reduced modulo the image $\mathrm{im}(d\colon C_1 \to C_0)$, i.e. the first components of level 0. No term of it is divisible by a leading term of level 0 from earlier degrees. **Outcome: new generator.** The relation $\mathcal{P}^1\iota$ is not a boundary, so C_1 gets a generator $[h_0]$ in degree 4 with $d[h_0] = \mathcal{P}^1\iota$. Level 0 gets the element $b_1^0 = (\mathcal{P}^1\iota \mid [h_0])$, with leading term $x_{01}\,\iota$. *End of degree 4.* New basis elements: level 0: b_1^0. New generators: $[h_0] \in C_1$.

Degree $t = 5$

The work of this degree: level 0: pair. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Level 0, pair (b_0^0, b_1^0). The leading terms $\mathrm{LT}(b_0^0) = Q_0\,\iota$ and $\mathrm{LT}(b_1^0) = x_{01}\,\iota$ have least common multiple $Q_0x_{01}\,\iota$, with cofactors x_{01} and Q_0. The leading terms of $\mathcal{P}^1 \cdot b_0^0$ and $Q_0 \cdot b_1^0$ are 1 and 1 times this monomial, so the S-element is $Q_0 \cdot b_1^0 - \mathcal{P}^1 \cdot b_0^0$:

$$(-Q_1\iota \mid Q_0[h_0] - \mathcal{P}^1[a_0]).$$

No term of the first component is divisible by a leading term of level 0. Multiply by -1 to make the leading coefficient 1. **Outcome: new basis element** $b_2^0 = (Q_1\iota \mid -Q_0[h_0] + \mathcal{P}^1[a_0])$ of level 0, with leading term $Q_1\,\iota$. *End of degree 5.* New basis elements: level 0: b_2^0. No new generators.

Degree $t = 6$

The work of this degree: level 0: pair. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Level 0, pair (b_0^0, b_2^0). The leading terms $\mathrm{LT}(b_0^0) = Q_0\,\iota$ and $\mathrm{LT}(b_2^0) = Q_1\,\iota$ have least common multiple $Q_0Q_1\,\iota$, with cofactors Q_1 and Q_0. The leading terms of $Q_1 \cdot b_0^0$ and $Q_0 \cdot b_2^0$ are -1 and 1 times this monomial, so the S-element is $-Q_0 \cdot b_2^0 - Q_1 \cdot b_0^0$:

$$(0 \mid -Q_0\mathcal{P}^1[a_0] - Q_1[a_0]).$$

The first component is zero, so the second component $z = -Q_0\mathcal{P}^1[a_0] - Q_1[a_0]$ is a cycle in C_1. Reduce it modulo $\mathrm{im}(d\colon C_2 \to C_1)$, the first components of level 1:

- The term $-Q_0\mathcal{P}^1[a_0]$ has leading monomial $Q_0x_{01}\,[a_0]$, divisible by $\mathrm{LT}(b_0^1)$ with quotient x_{01}. Adding $\mathcal{P}^1 \cdot \pi_1(b_0^1) = Q_0\mathcal{P}^1[a_0] + Q_1[a_0]$ cancels it and leaves 0.

Outcome: useless. Both components vanish, and the reduction contributed nothing. *End of degree 6.* No new generators.

Degree $t = 7$

There are no critical pairs, singles or relations in this degree. Nothing happens.

Degree $t = 8$

There are no critical pairs, singles or relations in this degree. Nothing happens.

Degree $t = 9$

The work of this degree: level 0: pair. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Level 0, pair (b_1^0, b_2^0). The leading terms $\mathrm{LT}(b_1^0) = x_{01}\,\iota$ and $\mathrm{LT}(b_2^0) = Q_1\,\iota$ have least common multiple $Q_1 x_{01}\,\iota$, with cofactors Q_1 and x_{01}. The leading terms of $Q_1 \cdot b_1^0$ and $\mathcal{P}^1 \cdot b_2^0$ are 1 and 1 times this monomial, so the S-element is $\mathcal{P}^1 \cdot b_2^0 - Q_1 \cdot b_1^0$:

$$(0 \mid -Q_0\mathcal{P}^1[h_0] + Q_1[h_0] - \mathcal{P}^2[a_0]).$$

The first component is zero, so the second component $z = -Q_0\mathcal{P}^1[h_0] + Q_1[h_0] - \mathcal{P}^2[a_0]$ is a cycle in C_1. No term of it is divisible by a leading term of level 1 from earlier degrees. The cycle $-Q_0\mathcal{P}^1[h_0] + Q_1[h_0] - \mathcal{P}^2[a_0]$ is set aside: cycles are tested once all results of level 0 in this degree are sorted.

The cycles of level 0. There is one. **Outcome: new generator.** The cycle $Q_0\mathcal{P}^1[h_0] - Q_1[h_0] + \mathcal{P}^2[a_0]$, multiplied by -1 to make the leading coefficient 1, is not a boundary, so C_2 gets a generator $[a_1]$ in degree 9 with $d[a_1] = Q_0\mathcal{P}^1[h_0] - Q_1[h_0] + \mathcal{P}^2[a_0]$. Level 1 gets the element $b_1^1 = (Q_0\mathcal{P}^1[h_0] - Q_1[h_0] + \mathcal{P}^2[a_0] \mid [a_1])$, with leading term $Q_0 x_{01}\,[h_0]$. *End of degree 9.* New basis elements: level 1: b_1^1. New generators: $[a_1] \in C_2$.

Degree $t = 10$

The work of this degree: level 0: single; level 1: single. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Level 1, single $Q_0 \cdot b_1^1$. In $E^0\mathcal{A}$ we have $Q_0^2 = 0$, so the leading term of $Q_0 \cdot b_1^1$ is not the product of the leading terms, and the product has to be reduced:

$$(-Q_0Q_1[h_0] + Q_0\mathcal{P}^2[a_0] \mid Q_0[a_1]).$$

Reduce the first component modulo level 1:

- The term $Q_0\mathcal{P}^2[a_0]$ has leading monomial $Q_0 x_{01}^2\,[a_0]$, divisible by $\mathrm{LT}(b_0^1)$ with quotient x_{01}^2. Adding $-\mathcal{P}^2 \cdot b_0^1 = (-Q_0\mathcal{P}^2[a_0] - Q_1\mathcal{P}^1[a_0] \mid -\mathcal{P}^2[a_0^2])$ cancels it and leaves $(-Q_0Q_1[h_0] - Q_1\mathcal{P}^1[a_0] \mid Q_0[a_1] - \mathcal{P}^2[a_0^2])$.

Multiply by -1 to make the leading coefficient 1. **Outcome: new basis element** $b_2^1 = (Q_0Q_1[h_0] + Q_1\mathcal{P}^1[a_0] \mid -Q_0[a_1] + \mathcal{P}^2[a_0^2])$ of level 1, with leading term $Q_0Q_1\,[h_0]$.

Level 0, single $Q_1 \cdot b_2^0$. In $E^0\mathcal{A}$ we have $Q_1^2 = 0$, so the leading term of $Q_1 \cdot b_2^0$ is not the product of the leading terms, and the product has to be reduced:

$$(0 \mid Q_0Q_1[h_0] + Q_1\mathcal{P}^1[a_0]).$$

The first component is zero, so the second component $z = Q_0Q_1[h_0] + Q_1\mathcal{P}^1[a_0]$ is a cycle in C_1. No term of it is divisible by a leading term of level 1 from earlier degrees. The second component contains the leading term of b_2^1, found in this degree; subtracting its first component leaves 0. **Outcome: useless.** The cycle is a multiple of the first component of b_2^1, found one level up earlier in this degree, so it is the boundary $d(-Q_0[a_1]+\mathcal{P}^2[a_0^2])$. Both components now vanish, and the reduction contributed nothing. *End of degree 10.* New basis elements: level 1: b_2^1. No new generators.

Degree $t = 11$

The work of this degree: level 1: single. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Level 1, single $Q_0 \cdot b_2^1$. In $E^0\mathcal{A}$ we have $Q_0^2 = 0$, so the leading term of $Q_0 \cdot b_2^1$ is not the product of the leading terms, and the product has to be reduced:

$$(Q_0Q_1\mathcal{P}^1[a_0] \mid Q_0\mathcal{P}^2[a_0^2]).$$

Reduce the first component modulo level 1:

- The term $Q_0Q_1\mathcal{P}^1[a_0]$ has leading monomial $Q_0Q_1x_{01}\,[a_0]$, divisible by $\mathrm{LT}(b_0^1)$ with quotient Q_1x_{01}. Adding $Q_1\mathcal{P}^1 \cdot b_0^1 = (-Q_0Q_1\mathcal{P}^1[a_0] \mid Q_1\mathcal{P}^1[a_0^2])$ cancels it and leaves $(0 \mid Q_0\mathcal{P}^2[a_0^2] + Q_1\mathcal{P}^1[a_0^2])$.

The first component is zero, so the second component $z = Q_0\mathcal{P}^2[a_0^2]+Q_1\mathcal{P}^1[a_0^2]$ is a cycle in C_2. Reduce it modulo $\mathrm{im}(d\colon C_3 \to C_2)$, the first components of level 2:

- The term $Q_0\mathcal{P}^2[a_0^2]$ has leading monomial $Q_0x_{01}^2\,[a_0^2]$, divisible by $\mathrm{LT}(b_0^2)$ with quotient x_{01}^2. Adding $-\mathcal{P}^2 \cdot \pi_1(b_0^2) = -Q_0\mathcal{P}^2[a_0^2] - Q_1\mathcal{P}^1[a_0^2]$ cancels it and leaves 0.

Outcome: useless. Both components vanish, and the reduction contributed nothing. *End of degree 11.* No new generators.

Degree $t = 12$

Relations of degree 12 would give generators of C_1 in stem $11 > 10$ and are no longer taken up. The work of this degree: level 0: single. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Level 0, single $x_{01}^2 \cdot b_1^0$. In $E^0\mathcal{A}$ we have $x_{01}^2 \cdot x_{01} = 0$ (as $x_{in}^3 = 0$), so the leading term of $\mathcal{P}^2 \cdot b_1^0$ is not the product of the leading terms, and the product has to be reduced:

$$(0 \mid \mathcal{P}^2[h_0]).$$

The first component is zero, so the second component $z = \mathcal{P}^2[h_0]$ is a cycle in C_1. No term of it is divisible by a leading term of level 1 from earlier degrees. The cycle $\mathcal{P}^2[h_0]$ is set aside: cycles are tested once all results of level 0 in this degree are sorted.

The cycles of level 0**.** There is one. **Outcome: new generator.** The cycle $\mathcal{P}^2[h_0]$ is not a boundary, so C_2 gets a generator $[b_0]$ in degree 12 with $d[b_0] = \mathcal{P}^2[h_0]$. Level 1 gets the element $b_3^1 = (\mathcal{P}^2[h_0] \mid [b_0])$, with leading term $x_{01}^2\,[h_0]$. *End of degree 12.* New basis elements: level 1: b_3^1. New generators: $[b_0] \in C_2$.

Degree $t = 13$

Level 0 is no longer needed: a cycle found at level s in this degree would be a generator of C_{s+2} in stem $13 - s - 2 > 10$, so its pairs are not formed. The work of this degree: level 1: pair. All of them are reduced independently (in parallel in the program), and the results are then sorted from the top level down.

Level 1**, pair** (b_1^1, b_3^1)**.** The leading terms $\mathrm{LT}(b_1^1) = Q_0 x_{01}\,[h_0]$ and $\mathrm{LT}(b_3^1) = x_{01}^2\,[h_0]$ have least common multiple $Q_0 x_{01}^2\,[h_0]$, with cofactors x_{01} and Q_0. The leading terms of $\mathcal{P}^1 \cdot b_1^1$ and $Q_0 \cdot b_3^1$ are -1 and 1 times this monomial, so the S-element is $-Q_0 \cdot b_3^1 - \mathcal{P}^1 \cdot b_1^1$:

$$(0 \mid \ -Q_0[b_0] - \mathcal{P}^1[a_1]).$$

The first component is zero, so the second component $z = -Q_0[b_0] - \mathcal{P}^1[a_1]$ is a cycle in C_2. No term of it is divisible by a leading term of level 2 from earlier degrees. The cycle $-Q_0[b_0] - \mathcal{P}^1[a_1]$ is set aside: cycles are tested once all results of level 1 in this degree are sorted.

The cycles of level 1**.** There is one. **Outcome: new generator.** The cycle $Q_0[b_0] + \mathcal{P}^1[a_1]$, multiplied by -1 to make the leading coefficient 1, is not a boundary, so C_3 gets a generator $[a_0 b_0]$ in degree 13 with $d[a_0 b_0] = Q_0[b_0] + \mathcal{P}^1[a_1]$. Level 2 gets the element $b_1^2 = (Q_0[b_0] + \mathcal{P}^1[a_1] \mid [a_0 b_0])$, with leading term $Q_0\,[b_0]$. *End of degree 13.* New basis elements: level 2: b_1^2. New generators: $[a_0 b_0] \in C_3$.

4 The result

Generators and differentials.

generator	s	t	$t-s$	differential
ι	0	0	0	—
$[a_0]$	1	1	0	$Q_0\iota$
$[a_0^2]$	2	2	0	$Q_0[a_0]$
$[a_0^3]$	3	3	0	$Q_0[a_0^2]$
$[h_0]$	1	4	3	$\mathcal{P}^1\iota$
$[a_1]$	2	9	7	$Q_0\mathcal{P}^1[h_0] - Q_1[h_0] + \mathcal{P}^2[a_0]$
$[b_0]$	2	12	10	$\mathcal{P}^2[h_0]$
$[a_0 b_0]$	3	13	10	$Q_0[b_0] + \mathcal{P}^1[a_1]$

Since the resolution is minimal, the generators are a basis of $\mathrm{Ext}_{\mathcal{A}}^{s,t}(\mathbb{F}_3, \mathbb{F}_3)$ in this range:

$s\backslash t-s$	0	1	2	3	4	5	6	7	8	9	10
3	a_0^3										a_0b_0
2	a_0^2							a_1			b_0
1	a_0			h_0							
0	1										

This is the familiar picture at $p = 3$. There is the a_0-tower in stem 0, h_0 in stem 3 (detecting α_1), a_1 in stem 7 (detecting α_2), and b_0 in stem 10 (detecting β_1). In a minimal resolution, the coefficient of Q_0 (respectively $\mathcal{P}^1$) on a generator g in d of a generator g' is the coefficient of g' in the Yoneda product $a_0 g$ (respectively $h_0 g$). So $d[a_0b_0] = Q_0[b_0] + \mathcal{P}^1[a_1]$ shows that the class in stem 10, $s = 3$, is $a_0b_0 = h_0a_1$, up to sign. The cycle beyond $s = 3$ found in degree 4 is a_0^4.

The Gröbner bases.

element	t	leading term	first component x_1	second component x_2
b_0^0	1	$Q_0\,\iota$	$Q_0\iota$	$[a_0]$
b_1^0	4	$x_{01}\,\iota$	$\mathcal{P}^1\iota$	$[h_0]$
b_2^0	5	$Q_1\,\iota$	$Q_1\iota$	$-Q_0[h_0] + \mathcal{P}^1[a_0]$
b_0^1	2	$Q_0\,[a_0]$	$Q_0[a_0]$	$[a_0^2]$
b_1^1	9	$Q_0x_{01}\,[h_0]$	$Q_0\mathcal{P}^1[h_0] - Q_1[h_0] + \mathcal{P}^2[a_0]$	$[a_1]$
b_2^1	10	$Q_0Q_1\,[h_0]$	$Q_0Q_1[h_0] + Q_1\mathcal{P}^1[a_0]$	$-Q_0[a_1] + \mathcal{P}^2[a_0^2]$
b_3^1	12	$x_{01}^2\,[h_0]$	$\mathcal{P}^2[h_0]$	$[b_0]$
b_0^2	3	$Q_0\,[a_0^2]$	$Q_0[a_0^2]$	$[a_0^3]$
b_1^2	13	$Q_0\,[b_0]$	$Q_0[b_0] + \mathcal{P}^1[a_1]$	$[a_0b_0]$

Counters. The statistics the program keeps per level ($s = -1$ are the module relations):

s	formed	divisible	chain	singles	reduced	useless	elements	boundary	gens	beyond	steps
-1	0	0	0	0	2	0	0	0	2	0	0
0	3	0	0	3	6	2	1	0	3	0	1
1	1	0	0	3	4	1	1	0	2	0	3
2	0	0	0	1	1	0	0	0	0	1	0
all	4	0	0	7	13	3	2	0	7	1	4

Here "formed" counts critical pairs, "divisible" and "chain" those removed by the Gebauer–Möller criteria (none in this small range), and "steps" the reduction steps. Of the 13 reductions, 3 were useless and one was beyond the range. The other 9 each produced a basis element or a generator.